

Probability and Statistics

Measures of Central Tendency

1. Arithmetic Mean (AM):

- **What it is:** Add all the numbers, then divide by how many numbers there are.
 - **Example:** If the numbers are 2, 4, 6,
 $AM = (2 + 4 + 6) / 3 = 4.$
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2. Geometric Mean (GM):

- **What it is:** Multiply all the numbers, then take the root (depending on how many numbers there are).
 - **Example:** If the numbers are 2, 4, 8,
 $GM = \sqrt[3]{(2 \times 4 \times 8)} = 4.$
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3. Harmonic Mean (HM):

- **What it is:** A kind of average that gives more importance to smaller numbers.
 - **Formula:** $HM = \text{Number of values} / \text{Sum of reciprocals of values}.$
 - **Example:** For 1, 2, 4,
 $HM = 3 / (1/1 + 1/2 + 1/4) = 1.71.$
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4. Median:

- **What it is:** The middle number when all the numbers are arranged in order.
- **Example:**
 - For 1, 3, 7: the median is **3**.
 - For 1, 3, 7, 9: the median is $(3 + 7) / 2 = 5.$

5. Mode:

- **What it is:** The number that appears the most often.
 - **Example:**
 - In 1, 2, 2, 3, the mode is **2**.
 - In 1, 1, 2, 2, 3, the modes are **1** and **2** (there are two modes).
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6. Weighted Mean:

- **What it is:** An average where some numbers are more important (have more "weight").
- **Example:** For numbers 2, 4, 6 with weights 1, 2, 3:
Weighted Mean = $[(2 \times 1) + (4 \times 2) + (6 \times 3)] / (1 + 2 + 3) = 4.67$.

Moments, Skewness, and Kurtosis in Statistics

1. Moments

Moments help describe the shape of a dataset. They are mathematical tools used to summarize different aspects of a distribution.

Types of Moments:

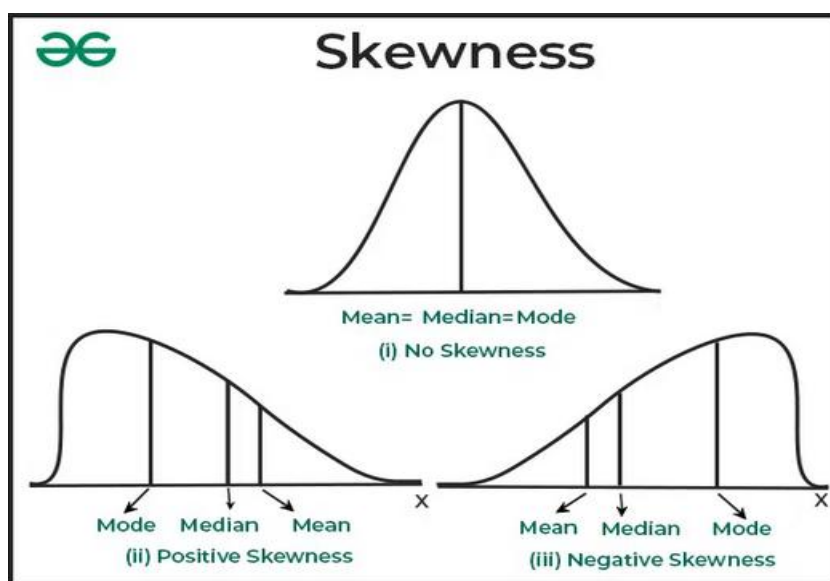
1. **First Moment (Mean):** This is the average of all data points.
 - Example: If your numbers are [1, 2, 3], the mean is calculated as:
2. **Second Moment (Variance):** This shows how spread out the data points are from the mean.
 - Example: In [1, 2, 3], the data isn't too spread out, so the variance is small.
3. **Third Moment (Skewness):** This measures the asymmetry of the data—whether it leans to one side.
4. **Fourth Moment (Kurtosis):** This measures the “tailedness” or extremity of the distribution—how fat or skinny the tails are.

2. Skewness

Skewness indicates whether the data distribution is symmetric or tilted to one side. It is calculated using the third moment (cubed deviations from the mean).

Types of Skewness:

- **No Skewness (0):** The distribution is perfectly balanced, like a mirror image on both sides.
- **Positive Skewness (>0):** The tail of the distribution stretches more to the right (it looks like a hump on the left).
- **Negative Skewness (<0):** The tail of the distribution stretches more to the left (it looks like a hump on the right).



Example: Imagine exam scores:

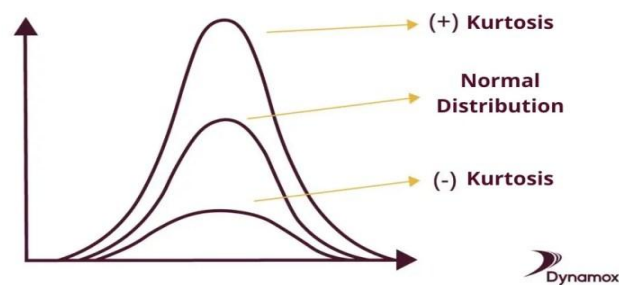
- Most students score low, but a few get very high marks—this would be **positively skewed**.

3. Kurtosis

Kurtosis measures the "peakedness" or "flatness" of a distribution. It tells us about the concentration of data in the tails.

Types of Kurtosis:

1. **Normal Kurtosis (3):** The distribution looks like a standard bell curve.
2. **High Kurtosis (>3):** The distribution has a steep peak and heavy tails, indicating more extreme values.
3. **Low Kurtosis (<3):** The distribution has a flatter peak and thinner tails, indicating fewer extreme values.



Example:

- **High Kurtosis:** Many people scoring either very low or very high in an exam.
- **Low Kurtosis:** Most people scoring around the same marks, like 50.

Quick Recap

- **Moments** describe the overall shape of the data.
 - First moment: Mean (location).
 - Second moment: Variance (spread).
 - Third moment: Skewness (tilt or asymmetry).
 - Fourth moment: Kurtosis (peakedness or tails).
- **Skewness** shows whether the data is balanced or tilted.
- **Kurtosis** shows whether the data has extreme values (heavy tails) or not.

Statistics

Probability: Probability is the measure of the likelihood that a specific event will occur, expressed as a value between 0 and 1, where 0 indicates impossibility and 1 indicates certainty.

Statistics: Statistics is the branch of mathematics concerned with collecting, analyzing, interpreting, presenting, and organizing data to make informed conclusions and decisions.

What are Measures of Variability?

Measures of variability, also called measures of dispersion, tell us how much the data values in a dataset are spread out or clustered around a central value like the mean or median. This helps us understand the consistency of data. For example, if a group of test scores has low variability, most students scored around the same amount. But if it has high variability, some students scored very high and others very low.

The most basic measure of variability is the **range**.

Variable

In statistics, a **variable** is any characteristic or value that can vary among people or items in a dataset. For example, height, age, or income are all variables because they can have different values for different people.

What are the Types of Variables?

Variables can be classified into different types based on the data they hold:

1. **Quantitative variables:** These have numerical values (like age or income).
2. **Qualitative (or categorical) variables:** These describe categories or qualities (like color or type of pet).
3. **Discrete variables:** Quantitative variables that can only take whole numbers (like the number of pets).
4. **Continuous variables:** Quantitative variables that can take any value within a range (like height or weight).

Why Do We Tend to Find the Center of Data?

We find the center of data (mean, median, mode) to understand what a "typical" value looks like in the dataset. For example, knowing the average income helps us understand the general financial situation in a community. But even if two datasets have the same mean, they can still look very different based on how spread out the data is around that center, which brings us back to variability.

Variability or Dispersion

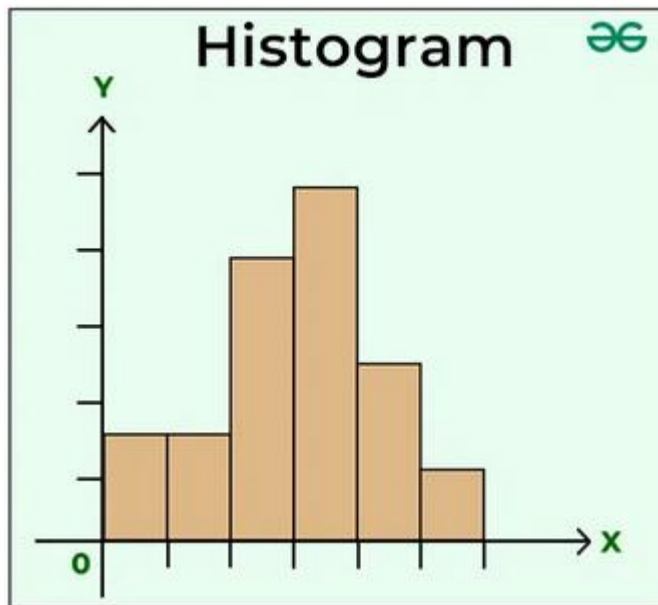
Variability (or dispersion) tells us how spread out the values in a dataset are. Understanding variability is important because it gives us insight into the consistency of data. For example, test scores with low variability mean students scored similarly, whereas high variability means scores were more spread out.

Key Measures of Variability

1. **Range:** The range is the simplest measure of variability. It's calculated by subtracting the smallest value in a dataset from the largest value. For example, if the highest birth weight in a dataset is 8 pounds and the lowest is 5 pounds, the range is 3 pounds.
 2. **Other measures:** The range is useful, but it only uses the extreme values in the data. Other measures (like variance and standard deviation) give more detailed information about the spread.
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1. Histogram

A **histogram** is a type of bar chart representing the distribution of numerical data. Each bar's height represents the frequency of data within a particular range (called a bin or interval). It's useful for visualizing the shape, spread, and center of the data distribution.



Steps to Determine Class Intervals (Bins) for a Histogram

To create a histogram, start by setting up class intervals (bins) to group your data. Here's how to find the ideal bin width and number of bins:

1. **Calculate the Data Range:** The range is simply the difference between the largest and smallest values:

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

For example, if your data goes from 5 to 50, then:

$$\text{Range} = 50 - 5 = 45$$

2. **Select the Number of Bins:** The number of bins, or classes, decides how many groups your data will be divided into. A common method for this is **Sturges' Rule**:

$$k = 1 + 3.322 \log_{10}(n)$$

Where:

- k = number of bins
- n = number of data points

For instance, if you have 100 data points:

$$k = 1 + 3.322 \log_{10}(100) = 1 + 3.322 \times 2 = 7.644 \approx 8 \text{ bins (rounded)}$$

Typically, 5–20 bins work well, depending on your data's distribution.

3. **Find the Class Width:** The class width is the range of values each bin will cover. Calculate it by dividing the range by the number of bins:

$$\text{Class Width} = \frac{\text{Range}}{\text{Number of Bins}}$$

For example, if the range is 45 and you decide on 8 bins:

$$\text{Class Width} = \frac{45}{8} = 5.625$$

Round this to a convenient number, like 6.

4. **Set Up Class Intervals:** Start with the minimum data value and add the class width to get each interval.

If the minimum value is 5 and the class width is 6, the intervals could look like this:

- 5–10
- 11–16
- 17–22
- 23–28
- etc.

2. Range

The **range** is the difference between the highest and lowest values in a data set. It gives a sense of the spread but is sensitive to extreme values:

Range = Maximum value – Minimum value

3. Average (Mean)

The **average** or **mean** is the sum of all data points divided by the number of points. It represents the central tendency of the data:

$$\text{Mean} = \frac{\sum \text{Data values}}{\text{Number of values}}$$

4. Median

The **median** is the middle value of a data set when arranged in ascending order. If there's an odd number of values, it's the central number; if even, it's the average of the two central numbers. The median is less affected by outliers than the mean.

5. Mode

The **mode** is the value that appears most frequently in a data set. There can be more than one mode (bimodal, multimodal) or no mode if all values are unique. The mode is useful for categorical data or data with repeating values.

6. Deviation:

In statistics, **deviation** is the difference between an individual data point and the mean of a dataset. It shows how much a value varies from the average.

Key Points:

- **Formula:**
Deviation = $x_i - \bar{x}$
where x_i is the data point and $\bar{x}$ is the mean.
- **Positive and Negative:** A positive deviation means the value is above the mean, while a negative deviation means it is below.
- **Relation to Variance and Standard Deviation:** Deviations are used to calculate variance (average of squared deviations) and standard deviation (square root of variance), which measure data spread.

Example:

For the dataset: 3, 5, 7, 9:

- Mean: $\bar{x}=6.25$
- Deviations:
 - 3: -3.25
 - 5: -1.25
 - 7: 0.75
 - 9: 2.75

These indicate how each number differs from the average.

6. Variance

Variance measures how far each value in the data set is from the mean. It shows the data's spread. The formula for variance for a population (σ^2) and sample (s^2) are:

$$\text{Population variance}(\sigma^2) = \frac{\sum (x_i - \mu)^2}{N}$$

$$\text{Sample variance}(s^2) = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

where μ is the population mean, $\bar{x}$ is the sample mean, and N and n are the number of values in the population and sample, respectively.

7. Standard Deviation

The **standard deviation** is the square root of variance. It also measures the data spread but in the same units as the original data:

$$\text{Standard Deviation} = \sqrt{\text{variance}}$$

It indicates the average distance from the mean and is often used in risk assessments and statistical analyses.

8. Tchebysheff's Theorem

Tchebysheff's Theorem (also known as **Chebyshev's Inequality**) is a statistical theorem that helps estimate the proportion of observations within a certain number of standard deviations from the mean for any probability distribution.

Key Points

- **Statement:** For any real-valued random variable with a finite mean μ and finite non-zero variance σ^2 , the proportion of observations that lie within k standard deviations from the mean is at least:

$$1 - (1/k^2)$$

for $k > 1$.

- **Interpretation:** This means that:
 - At least 75% of the data will fall within 2 standard deviations from the mean ($k=2$).

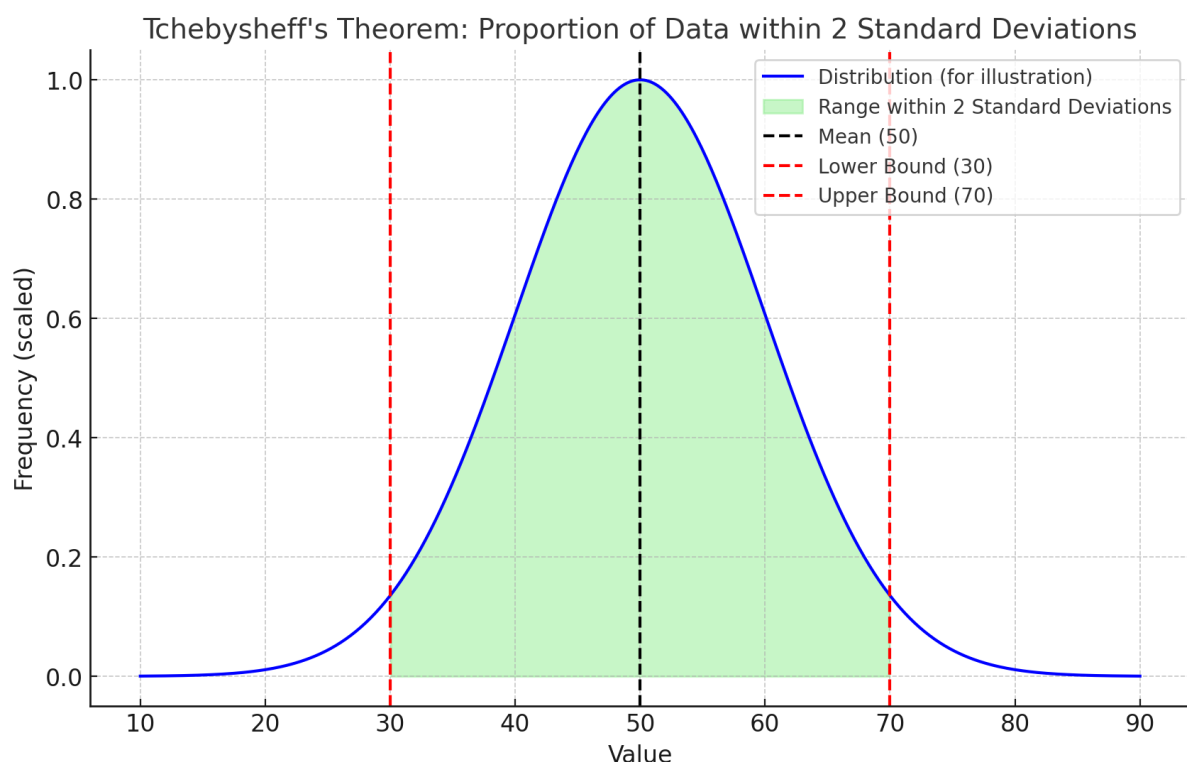
- At least 89% will fall within 3 standard deviations from the mean ($k=3$).
- At least 94% will fall within 4 standard deviations from the mean ($k=4$).
- **Applicability:** Tchebysheff's Theorem is applicable to **all distributions, regardless of shape**, making it especially useful when the distribution is unknown.

Example

If a dataset has a mean of 50 and a standard deviation of 10:

- For $k=2$:
 - At least $1 - (1/2^2) = 0.75$ or 75% of the data will lie within 50 ± 20 (i.e., between 30 and 70).

This theorem is useful for understanding the spread of data and making inferences about data distributions without specific distributional assumptions.



Here's a deviation graph illustrating the dataset with a mean of 50 and a standard deviation of 10. The shaded green area represents the range within 2 standard deviations (from 30 to 70), where Tchebysheff's Theorem guarantees that at least 75% of the data will fall within this interval.

Comprehensive Notes on Probability

Introduction to Probability

Probability is a mathematical tool used to evaluate how likely an event is to occur. It is used to make conclusions about a population based on sample data. For example, if we toss a fair coin many times, we expect the number of heads (H) and tails (T) to be nearly equal. However, if we get 10 heads in a row, this would suggest that the coin is biased.

Key Terminology

1. Experiment

An experiment refers to the process of observing or measuring something in a controlled environment. Examples include:

- Recording test grades
- Measuring rainfall
- Testing whether a circuit board is defective or not
- Tossing a coin and recording the result

2. Simple Event

A simple event is the outcome of a single trial of an experiment. For example, when tossing a die, the simple events are the results of a single toss:

- E_1 : Observe 1
- E_2 : Observe 2
- E_3 : Observe 3
- E_4 : Observe 4
- E_5 : Observe 5
- E_6 : Observe 6

3. Event

An event is a collection of simple events. For example, consider the event of observing an odd number when tossing a die:

- A: Observe an odd number, where $A = \{E_1, E_3, E_5\}$

Similarly, the event of observing a number less than 4 is:

- B: Observe a number less than 4, where $B = \{E_1, E_2, E_3\}$

4. Mutually Exclusive Events

Two events are mutually exclusive if they cannot happen at the same time. For example, in the die toss experiment, the events A (odd number) and B (less than 4) are mutually exclusive because the outcomes 1, 3, and 5 are the only odd numbers, and the outcomes 1, 2, and 3 are the numbers less than 4. But no outcome can be both odd and less than 4 simultaneously.

5. Sample Space

The sample space (S) is the set of all possible outcomes of an experiment. For a single coin toss, the sample space is:

$$- S = \{H, T\}$$

If you toss the coin twice, the sample space expands to include all possible combinations of heads and tails:

$$- S = \{HH, HT, TH, TT\}$$

For a die toss, the sample space is:

$$- S = \{1, 2, 3, 4, 5, 6\}$$

6. Relative Frequency

The relative frequency of an event is the ratio of the number of times the event occurs to the total number of trials in an experiment. It is calculated as:

$$- \text{Relative frequency} = \text{Frequency of event A} / n$$

Where n is the total number of trials.

As the number of trials n approaches infinity, the relative frequency of an event tends to stabilize and approaches the probability of that event.

7. Probability

The probability of an event is defined as the relative frequency of that event occurring in a large number of trials. It lies between 0 and 1:

- $P(A) = 0$ means the event A never occurs.

- $P(A) = 1$ means the event A always occurs.

- $P(A)$ can be any value between 0 and 1, indicating the likelihood of event A happening.

For example, when tossing a fair six-sided die, the probability of rolling any specific number (like 1, 2, 3, 4, 5, or 6) is:

$$- P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$$

This is because each outcome has an equal chance of occurring.

8. Requirements for Simple Event Probabilities

1. Each probability value must lie between 0 and 1:
 - $0 \leq P(A) \leq 1$
2. The sum of the probabilities of all simple events in the sample space must equal 1:
 - $P(S) = 1$

9. Probability of an Event

The probability of an event A is the sum of the probabilities of the simple events that make up the event:

- $P(A) = \sum P(\text{simple events in } A)$

For example, if you toss two coins, the event A of observing exactly one head (H) would consist of the simple events HT and TH. So:

- $P(A) = P(HT) + P(TH)$

10. Example: Tossing Two Coins

Let's calculate the probability of observing exactly one head when tossing two fair coins.

- The sample space for two coin tosses is:
- $S = \{HH, HT, TH, TT\}$
- The event of observing exactly one head is:
- $A = \{HT, TH\}$
- The probability of each outcome (since the coins are fair) is $1/4$.

Therefore, the probability of event A is:

- $P(A) = P(HT) + P(TH) = 1/4 + 1/4 = 2/4 = 1/2$

Event Relations and Probability Rules

Introduction to Event Relations

Sometimes the event of interest can be formed as a combination of several other events. In these cases, we need to consider three important event rules, which are similar to set theory.

1. The Union Rule

The union of events A and B, denoted by $A \cup B$, is the event that either A or B or both occur.

2. The Intersection Rule

The intersection of events A and B, denoted by $A \cap B$, is the event that both A and B occur.

3. The Complement Rule

The complement of an event A, denoted by A^c (or A'), is the event that A does not occur.

$$A^c = 1 - A$$

Example: Two Fair Coin Tosses

Consider two fair coins being tossed, and the outcome is recorded. The following are the events of interest:

- A: Observe at least one head.
- B: Observe at least one tail.

Define the events A, B, $A \cup B$, $A \cap B$, and A^c as collections of simple events, and find their probabilities.

Calculating Probabilities for Unions and Complements

4. The Addition Rule

Given two events, A and B, the probability of their union, $A \cup B$, is given by:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

5. Mutually Exclusive Events

If two events are mutually exclusive, meaning that they cannot occur at the same time, then:

$$P(A \cup B) = P(A) + P(B)$$

6. Rule for Complement

The probability of the complement of an event A, denoted by A^c , is:

$$P(A) + P(A^c) = 1$$

Or equivalently:

$$P(A^c) = 1 - P(A)$$

Independence, Conditional Probability, and The Multiplication Rule

7. Independent and Dependent Events

There is a probability rule that can be used to calculate the probability of the intersection of several events. However, this rule depends on the important statistical concept of independent or dependent events.

8. Definition of Independent Events

Two events, A and B, are said to be independent if and only if the probability of event B is not influenced or changed by the occurrence of event A, or vice versa.

9. General Multiplicative Rule

The probability that both events A and B occur when the experiment is performed is:

$$P(A \cap B) = P(A) * P(B|A)$$

10. Conditional Probability

The probability of event A, given that event B has occurred, is called the conditional probability of A, given that B has occurred, denoted by $P(A|B)$.

The formula for conditional probability is:

$$P(A|B) = P(A \cap B) / P(B), \text{ given that } P(B) \neq 0$$

Similarly, the conditional probability of B given A is:

$$P(B|A) = P(A \cap B) / P(A), \text{ given that } P(A) \neq 0$$

Random Variables, Mathematical Expectation, and Variance

1. Random Variables

A random variable (RV) is a variable whose value is determined by the outcome of a random experiment or process. It can take on different values depending on the result of the experiment.

Example:

- Tossing a die and recording the number that appears on the upper face: Here, the random variable x can take values from 1 to 6.
- Number of defects in a randomly selected piece of furniture.
- SAT score of a randomly selected college applicant.
- Number of telephone calls received by a crisis intervention hotline during a selected time period.

Types of Random Variables

- **Discrete Random Variable:** A random variable that can only take a finite or countable number of distinct values (e.g., number of heads in a coin toss, number of defects in a batch).
- **Continuous Random Variable:** A random variable that can take any value within a certain range, representing measurements that can be infinitely divided (e.g., height, weight, time).

2. Qualitative vs Quantitative Variables

- **Qualitative Variables:** Variables that describe categories or qualities (e.g., gender, color, types of animals).
- **Quantitative Variables:** Variables that can be counted or measured (e.g., number of defects, time taken to complete a task, income level).

Identifying Qualitative and Quantitative Variables:

1. The most frequent use of your microwave oven: Qualitative (categorical: reheating, defrosting, etc.)
2. The number of consumers who refuse to answer a telephone survey: Quantitative (countable number of consumers).

3. The door chosen by a mouse in a maze experiment: Qualitative (categorical: A, B, or C).
4. The winning time for a horse running in the Kentucky Derby: Quantitative (measurable time value).
5. The number of children in a fifth-grade class reading at or above grade level: Quantitative (countable number of children).

3. Probability Distributions

A probability distribution defines the probabilities of each possible outcome for a random variable.

Discrete Probability Distribution

For a discrete random variable, it is a list or function that associates each possible value of the random variable with a probability. The sum of all probabilities must equal 1.

Requirements for a Discrete Probability Distribution:

1. The probability for each value $p(x)$ must be between 0 and 1:
 $0 \leq p(x) \leq 1$
2. The sum of all probabilities must be equal to 1:
 $\sum p(x) = 1$

Example: Tossing Two Fair Coins

Let x be the number of heads observed. The possible values for x are 0, 1, and 2.

- $p(x = 0) = 1/4$ (No heads, TT)
- $p(x = 1) = 1/2$ (One head, HT, TH)
- $p(x = 2) = 1/4$ (Two heads, HH)

Thus, the probability distribution for x is:

- $P(x = 0) = 1/4$
- $P(x = 1) = 1/2$
- $P(x = 2) = 1/4$

4. Mathematical Expectation (Mean) of a Random Variable

The expected value or mean of a random variable x , denoted μ , is a measure of the central tendency of the random variable. It represents the average value of x over many trials.

Formula:

For a discrete random variable x with probability distribution $p(x)$, the expected value μ is calculated as:

$$\mu = E(x) = \sum xp(x)$$

Where:

- x is a possible value of the random variable x ,
- $p(x)$ is the probability associated with x .

Example: Die Toss

For the die toss, where x = the number on the die (with probabilities $p(x) = 1/6$ for each outcome):

$$\mu = E(x) = 1 * 1/6 + 2 * 1/6 + 3 * 1/6 + 4 * 1/6 + 5 * 1/6 + 6 * 1/6 = 3.5$$

Thus, the expected value of x is 3.5.

5. Variance and Standard Deviation

- Variance measures how much the values of the random variable deviate from the expected value (mean).
- Standard Deviation is the square root of the variance and gives a measure of the spread of the values in the same units as the random variable.

Formula for Variance:

The variance σ^2 of a discrete random variable x with mean μ is given by:

$$\sigma^2 = E[(x - \mu)^2] = \sum (x - \mu)^2 p(x)$$

Formula for Standard Deviation:

The standard deviation σ is the square root of the variance:

$$\sigma = \sqrt{\text{Var}(x)}$$

Example: Die Toss

For the die toss, where the mean $\mu = 3.5$:

The variance σ^2 is:

$$\sigma^2 = (1 - 3.5)^2 * 1/6 + (2 - 3.5)^2 * 1/6 + (3 - 3.5)^2 * 1/6 + (4 - 3.5)^2 * 1/6 + (5 - 3.5)^2 * 1/6 + (6 - 3.5)^2 * 1/6$$

After calculating, the variance is approximately 2.9167.

The standard deviation $\sigma = \sqrt{2.9167} \approx 1.707$.

6. Summary of Key Concepts

- A random variable is a variable whose value depends on the outcome of a random experiment.
- Discrete random variables take countable values, while continuous random variables take values within a range.
- The mean (expected value) of a random variable represents its average outcome.
- The variance measures the spread of the values around the mean, and the standard deviation is the square root of the variance.

Bayes' Theorem

1. Introduction to Bayes' Theorem

Bayes' Theorem is a fundamental concept in probability theory and statistics that describes the probability of an event, based on prior knowledge of conditions that might be related to the event. It is named after Reverend Thomas Bayes, who introduced the concept in the 18th century.

Bayes' Theorem allows you to update the probability of a hypothesis (event) as more evidence or data becomes available. It is particularly useful in situations where we have incomplete information, and we want to revise our beliefs about the probability of events based on new data.

2. The Formula of Bayes' Theorem

Bayes' Theorem can be expressed as:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:

- **$P(A|B)$** is the **posterior probability**: the probability of event **A** occurring given that **B** has occurred.
- **$P(B|A)$** is the **likelihood**: the probability of event **B** occurring given that **A** is true.
- **$P(A)$** is the **prior probability**: the initial probability of event **A** before any evidence (B) is taken into account.

- **$P(B)$** is the **marginal likelihood** or **evidence**: the total probability of event **B** occurring, considering all possible scenarios.

Understanding Bayes' Theorem

You feel unwell and consult a doctor. The doctor conducts a test, and the result is positive for a rare disease that affects 0.1% of the population. You ask, "How certain are you that I have the disease?" The test has the following characteristics:

- **Prevalence of the disease ($P(A)$):** $0.1\% = 0.001$
- **True Positive Rate ($P(B|A)$):** $99\% = 0.99$ (sensitivity)
- **False Positive Rate ($P(B|\neg A)$):** $1\% = 0.01$

The task is to calculate the probability that you actually have the disease, given a positive test result ($P(A|B)$).

To calculate this:

- You need the probability of testing positive if you have the disease ($P(B|A)$).
- The overall probability of having the disease ($P(A)$).
- The overall probability of testing positive ($P(B)$).

Components of Bayes' Theorem

1. Prior Probability ($P(A)$):

The initial belief about an event before considering new data.

Example: If 1% of people have a disease, **$P(A) = 0.01$** .

2. Likelihood ($P(B|A)$):

The probability of observing evidence (B) if the event (A) is true.

Example: If 95% of people with the disease test positive, **$P(B|A) = 0.95$** .

3. Marginal Likelihood (P(B)):

The total probability of the evidence, considering all cases.

Example: The probability of testing positive, including healthy and diseased people.

4. Posterior Probability (P(A|B)):

The updated probability of the event (A) after considering the evidence (B).

Bayes' Theorem Example: Medical Test

- Prior Probability (P(A)): 1% of people have a disease. $P(A) = 0.01$.
- Likelihood (P(B|A)): The test correctly identifies the disease 95% of the time. $P(B|A) = 0.95$.
- False Positive Rate: 5% of healthy people test positive.
- Marginal Likelihood (P(B)):

$$P(B) = P(B|A) \cdot P(A) + P(B| - A) \cdot P(-A)$$

Substitute values:

$$P(B) = 0.95 \cdot 0.01 + 0.05 \cdot 0.99 = 0.059$$

- Posterior Probability (P(A|B)):

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} = \frac{0.95 \cdot 0.01}{0.059} \approx 0.161$$

This means if you test positive, there's about a **16.1% chance** you actually have the disease.

A Hypothetical Scenario

You test positive for a very rare disease affecting 0.1% of the population.

- Prior probability of the disease: $P(H) = 0.001$.
- Test sensitivity (correctly identifies disease): $P(E|H) = 0.99$.
- Test specificity (correctly identifies healthy): $P(E|-H) = 0.99$.

Using Bayes' Theorem:

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E|H) \cdot P(H) + P(E|-H) \cdot P(-H)}$$

Substitute values:

$$P(H|E) = \frac{0.99 \cdot 0.001}{0.99 \cdot 0.001 + 0.01 \cdot 0.999} \approx 0.09$$

Even after testing positive, there's only a **9% chance** of having the disease.

Second Opinion

If you test positive again from another doctor, Bayes' Theorem can be applied again to update the probability further based on this new evidence.

6. Generalization to Multiple Events

Bayes' Theorem can also be extended to more than two events. For multiple hypotheses, it is expressed as:

$$P(A_i|B) = \frac{P(B|A_i) \cdot P(A_i)}{\sum_i P(B|A_i) \cdot P(A_i)}$$

Where $A_1, A_2, \dots$ are different hypotheses, and the denominator sums over all possible hypotheses.

Discrete Probability Distribution

What is Discrete Probability Distribution?

A discrete probability distribution describes the likelihood of outcomes for a discrete random variable. A random variable is called discrete if it takes on finite or countable values. The distribution gives the probability of each possible value that the random variable can take.

Conditions for Discrete Probability Distribution

1. Probability Values: The probability of each outcome must lie between 0 and 1.

$$0 \leq P(X=x) \leq 1$$

2. Sum of Probabilities: The sum of all probabilities must equal 1.

$$\sum P(X=x) = 1$$

Discrete Probability Distribution Example

Consider tossing two coins. Let X represent the number of tails:

Sample Space: {HH, HT, TH, TT}

Values of X (Number of Tails):

X=0 (HH): Probability $P(X=0) = 1/4$

X=1 (HT, TH): Probability $P(X=1) = 1/2$

X=2 (TT): Probability $P(X=2) = 1/4$

Discrete Probability Distribution Formulas

1. **Probability Mass Function (PMF):** PMF of a discrete random variable X is the value completely equal to x.

$$f(x) = P(X=x)$$

What is PMF?

- PMF tells you the chance (probability) of each specific value of a discrete random variable happening.
- A discrete random variable means something that can only take specific values (e.g., rolling a die gives 1, 2, 3, 4, 5, or 6).

Imagine rolling a fair die:

- Possible outcomes are 1,2,3,4,5,6.
- The PMF gives the probability of getting each number.
- Since the die is fair, each number has an equal chance:

$$P(1) = \frac{1}{6}, P(2) = \frac{1}{6}, \dots, P(6) = \frac{1}{6}.$$

2. Cumulative Distribution Function (CDF): CDF of a discrete random variable X is less than or equal to value x . In simpler terms, it's the total probability up to a certain value.

$$F(x) = P(X \leq x)$$

Example: Rolling a Die

- X is the outcome of rolling a fair die (1, 2, 3, 4, 5, 6):

x	$P(X = x)$ (PMF)	$F(x) = P(X \leq x)$ (CDF)
1	$\frac{1}{6}$	$\frac{1}{6}$
2	$\frac{1}{6}$	$\frac{2}{6}$
3	$\frac{1}{6}$	$\frac{3}{6}$
4	$\frac{1}{6}$	$\frac{4}{6}$
5	$\frac{1}{6}$	$\frac{5}{6}$
6	$\frac{1}{6}$	1

3. Mean (Expected Value):

$$E[X] = \sum x \cdot P(X=x)$$

4. Variance:

$$\text{Var}[X] = \sum (x - \mu)^2 P(X = x)$$

where $\mu = E[X]$

Types of Discrete Probability Distributions

1. Bernoulli Distribution:

Used for random experiments with two outcomes (success/failure).

PMF:

$P(X=1) = p$ (probability of success)

$P(X=0) = 1-p$ (probability of failure)

2. Binomial Distribution:

Describes the number of successes in a fixed number of independent trials.

PMF:

$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$, where n is the number of trials, p is the probability of success, and x is the number of successes.

3. Poisson Distribution:

Describes the probability of a given number of events occurring in a fixed interval of time or space.

PMF:

$P(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}$, where λ is the average number of events in the interval.

4. Geometric Distribution:

Describes the number of trials until the first success.

PMF:

$P(X=x) = (1-p)^{x-1} p$

How to Find a Discrete Probability Distribution

Step 1: Determine the sample space of the experiment.

Step 2: Define the random variable X .

Step 3: Identify all possible values of X and their associated probabilities.

Step 4: Organize the values and probabilities into a table.

Examples

Example 1: Tossing two coins, find the discrete probability distribution for the number of heads:

Sample Space: {HH, HT, TH, TT}

X (Number of heads):

$X=0$ (TT): Probability $1/4$

$X=1$ (HT, TH): Probability $1/2$

$X=2$ (HH): Probability $1/4$

Example 2: Given the probability table:

$$P(X=3) = 0.1, P(X=4) = p, P(X=5) = 0.2, P(X=6) = 0.4$$

Find p :

By the sum of probabilities condition:

$$0.1 + p + 0.2 + 0.4 = 1$$

Solving:

$$p = 0.3$$

Example 3: Find the mean of a discrete distribution:

Given:

$$X = 2, 3, 4, 5$$

$$P(X=2) = 0.16, P(X=3) = 0.45, P(X=4) = 0.32, P(X=5) = 0.07$$

Mean:

$$E[X] = 2 \times 0.16 + 3 \times 0.45 + 4 \times 0.32 + 5 \times 0.07 = 3.3$$

Practice Problems

1. Construct the discrete probability table when a die is rolled and X represents the numbers greater than or equal to 3.

2. Find a from the following probability table:

$$X = 0, 1, 2, 4$$

$$P(X=0) = 0.6, P(X=1) = a, P(X=2) = 0.1, P(X=4) = 0.3$$

3. Find the expected value of the following distribution:

$$X = 4, 5, 6, 7$$

$$P(X=4) = 0.21, P(X=5) = 0.35, P(X=6) = 0.42, P(X=7) = 0.02$$

FAQs

1. What is a discrete probability distribution?

It describes the probabilities of a discrete random variable.

2. What are the requirements of a discrete probability distribution?

The probability of each value must be between 0 and 1, and the sum of all probabilities must be 1.

3. Can the expected value of a discrete probability distribution be negative?

Yes, it can be negative depending on the distribution's values.

Binomial Distribution

Definition

The binomial distribution describes the number of successes in n independent trials, where each trial has two possible outcomes: success or failure.

Key Features

1. **Fixed Number of Trials (n):** The experiment is repeated n times.
2. **Two Outcomes:** Each trial has success (p) or failure ($1-p$).
3. **Independent Trials:** The outcome of one trial doesn't affect another.
4. **Constant Probability (p):** The probability of success is the same for all trials.

Formula

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Where:

X : Number of successes

n : Total trials

k : Number of successes

p : Probability of success

$\binom{n}{k} = \frac{n!}{k!(n-k)!}$: Binomial coefficient.

Examples

1. Coin Toss

Problem Statement:

You toss a fair coin $n = 3$ times. What is the probability of getting exactly $k = 2$ heads, assuming the probability of heads (p) is 0.5?

Solution:

$$P(X = 2) = \binom{3}{2} (0.5)^2 (0.5)^1 = 3 \cdot 0.25 \cdot 0.5 = 0.375$$

Answer: $P(X = 2) = 0.375$ or 37.5%.

2. Basketball Player

Problem Statement:

A basketball player has a free throw success rate of $p = 0.8$. He attempts $n = 4$ free throws:

- a. What is the probability of making exactly $k = 2$ successful free throws?
- b. What is the probability of making at least one successful free throw ($X \geq 1$)?

Solution (a):

$$P(X = 2) = \binom{4}{2} (0.8)^2 (0.2)^2 = 6 \cdot 0.64 \cdot 0.04 = 0.1536$$

Answer: $P(X = 2) = 0.1536$ or 15.36%.

Solution (b):

$$P(X \geq 1) = 1 - P(X = 0)$$

First, find $P(X = 0)$:

$$P(X = 0) = \binom{4}{0} (0.8)^0 (0.2)^4 = 1 \cdot 1 \cdot 0.0016 = 0.0016$$

Now:

$$P(X \geq 1) = 1 - 0.0016 = 0.9984$$

Answer: $P(X \geq 1) = 0.9984$ or 99.84%.

Applications

1. Quality Control: Proportion of defective products in a batch.
2. Surveys: Proportion of 'yes' answers in a questionnaire.
3. Biology: Proportion of individuals with a genetic trait.

Binomial Random Variables

Random variables that count the number of successes in a binomial experiment.

Examples:

- Proportion of teachers who are men in a survey.
- Number of cola drinkers preferring a certain brand.
- Proportion of people with a genetic marker.

Rule of Thumb

If $n/N \geq 0.05$ (sample size is large compared to population size), the binomial assumption may not hold.

Poisson Distribution

Definition

The Poisson distribution models the number of occurrences of an event in a fixed interval of time, space, or other unit, where:

- Events occur independently of each other.
- The average rate (μ) of events is constant within the interval.
- Two events cannot occur simultaneously.

Key Features

1. Discrete Random Variable: Counts occurrences of an event.
2. Parameter (μ): Represents the average rate of events.
3. Independence: Events do not influence one another.
4. Random Events: Events occur randomly within the interval.

Formula

The probability of observing k events in a given interval:

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!}$$

Where:

- X: Number of events
- k: Specific number of occurrences
- μ : Average rate of occurrences
- e: Euler's number (≈ 2.718)
- $k!$: Factorial of k .

Mean and Standard Deviation

Mean: μ

Standard Deviation: $\sqrt{\mu}$

Examples of Applications

1. Number of calls received by a switchboard in a given time.
2. Number of bacteria in a specific fluid volume.
3. Customer arrivals at a checkout counter per minute.
4. Machine breakdowns in a day.
5. Traffic accidents at an intersection during a specific time.

Examples

1. Traffic Accidents (1 Week)

Problem Statement:

The average number of traffic accidents on a highway is $\mu = 2$ per week.

- Find the probability of **no accidents** during a 1-week period.

Solution:

Using the formula:

$$P(X = 0) = \frac{\mu^0 e^{-\mu}}{0!} = e^{-2}$$

Approximating $e^{-2} \approx 0.135$:

$$P(X = 0) = 0.135$$

Answer: $P(X = 0) = 0.135$ or 13.5%.

2. Traffic Accidents (2 Weeks)

Problem Statement:

For the same highway, find the probability of **at most 3 accidents** during a 2-week period ($\mu=4$).

Solution:

Here, $\mu = 4$, and we need to calculate $P(X \leq 3)$:

$$P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

Calculate each term:

$$P(X = 0) = \frac{4^0 e^{-4}}{0!} = e^{-4} \approx 0.0183$$

$$P(X = 1) = \frac{4^1 e^{-4}}{1!} = 4 \cdot e^{-4} \approx 0.0733$$

$$P(X = 2) = \frac{4^2 e^{-4}}{2!} = 8 \cdot e^{-4} \approx 0.1465$$

$$P(X = 3) = \frac{4^3 e^{-4}}{3!} = \frac{64}{6} \cdot e^{-4} \approx 0.1954$$

Add these probabilities:

$$P(X \leq 3) = 0.0183 + 0.0733 + 0.1465 + 0.1954 = 0.4335$$

Answer: $P(X \leq 3) = 0.4335$ or **43.35%**.

Poisson as an Approximation of Binomial

The Poisson distribution provides an approximation for binomial probabilities when:

1. The number of trials (n) is large.
2. The probability of success (p) is small.
3. The mean $\mu=np$ is used in place of n and p .

Continuous Probability Distribution

A **continuous probability distribution** describes situations where the random variable can take any value within a certain range. Unlike discrete distributions (where outcomes are specific values), continuous distributions have infinite possible outcomes within a range.

Key Points:

- **Infinite Possibilities:** A continuous random variable can take any value in an interval (like between 0 and 1).
- **Probability Density Function (PDF):** Describes the likelihood of the variable taking a specific value.
- **Area Under the Curve:** The probability of the variable being in a range is given by the area under the curve of the PDF.
- **Cumulative Distribution Function (CDF):** Gives the probability that the random variable is less than or equal to a certain value.

Common Continuous Distributions:

Uniform Distribution

A uniform distribution is a type of probability distribution where every outcome within a given range is equally likely to occur. This means that all outcomes in the range have the same probability.

Formula:

The probability density function (PDF) for a continuous uniform distribution is given by:

$$f(x) = \frac{1}{b - a} \quad \text{for } a \leq x \leq b$$

Where:

- a is the **lower limit** of the range.
- b is the **upper limit** of the range.
- $f(x)$ is the probability density for any value between a and b .

This means that the height of the curve (probability) is constant between a and b.

Example:

Imagine you roll a fair die, which has 6 faces. Each number (1, 2, 3, 4, 5, 6) is equally likely to appear. This is a simple example of a uniform distribution.

- The possible outcomes are: {1, 2, 3, 4, 5, 6}.
- The probability of rolling any specific number (say, rolling a 4) is the same for all numbers, so it's a uniform distribution.

If we treat this situation as a continuous distribution, the range would be from $a = 1$ to $b = 6$. The formula becomes:

$$f(x) = \frac{1}{6 - 1} = \frac{1}{5} \quad \text{for } 1 \leq x \leq 6$$

This means that any value between 1 and 6 (if we look at it as a continuous range) has a probability density of $1/5$.

Normal Distribution

Normal distribution is a type of probability distribution that looks like a bell curve. It's also known as the Gaussian distribution. A normal distribution is a way to describe how data is spread out. It's like a bell-shaped curve where:

- Most of the data points are near the middle (average or mean).
- As you move away from the middle, there are fewer data points.
- The curve is symmetrical, meaning the left side is just like the right side.

Formula:

The probability density function (PDF) for a normal distribution is given by:

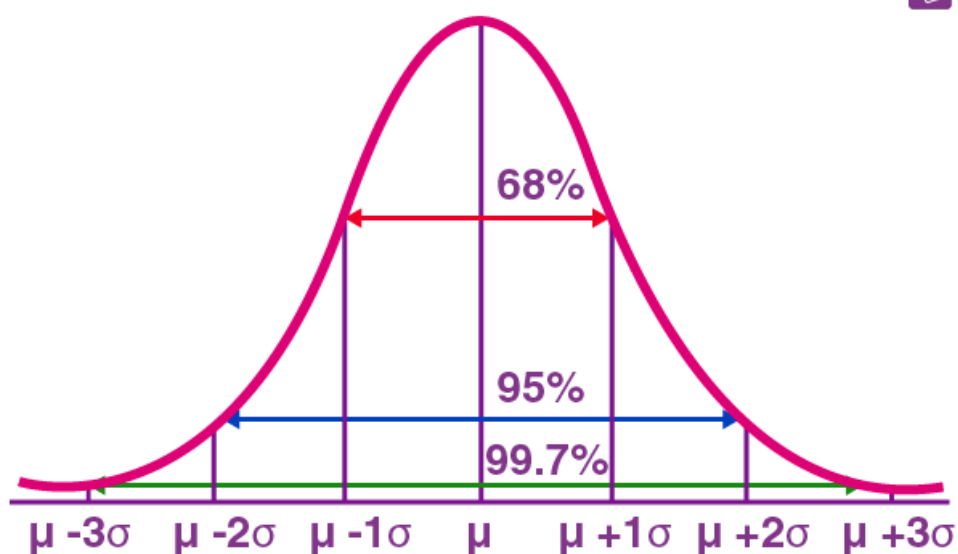
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

- μ = mean of the distribution
- σ = standard deviation
- $\pi \approx 3.14159$
- $e \approx 2.718$ (Euler's number)

Key Characteristics:

- **Symmetry:** The curve is symmetric around the mean.
- **Bell-shaped curve:** The distribution has a peak at the mean and falls off as values move further from the mean.
- **68-95-99.7 rule:**
 - About 68% of values lie within one standard deviation of the mean.
 - About 95% lie within two standard deviations.
 - About 99.7% lie within three standard deviations.



Normal Distribution Problems and Solutions

Question 1: Calculate the probability density function of normal distribution using the following data. $x = 3$, $\mu = 4$ and $\sigma = 2$.

Solution: Given, variable, $x = 3$

Mean = 4 and

Standard deviation = 2

By the formula of the probability density of normal distribution, we can write;

$$f(3, 4, 2) = \frac{1}{2\sqrt{2\pi}} e^{\frac{-(3-4)^2}{2 \times 2^2}}$$

Hence, $f(3,4,2) = 1.106$.

Question 2: If the value of random variable is 2, mean is 5 and the standard deviation is 4, then find the probability density function of the gaussian distribution.

Solution: Given,

Variable, $x = 2$

Mean = 5 and

Standard deviation = 4

By the formula of the probability density of normal distribution, we can write;

$$f(2, 5, 4) = \frac{1}{4\sqrt{2\pi}} e^{\frac{-(2-5)^2}{2 \times 4^2}}$$

$$f(2,5,4) = 1/(4\sqrt{2\pi}) e^0$$

$$f(2,5,4) = 0.0997$$

There are two main parameters of normal distribution in statistics namely mean and standard deviation. The location and scale parameters of the given normal distribution can be estimated using these two parameters.

Key Points:

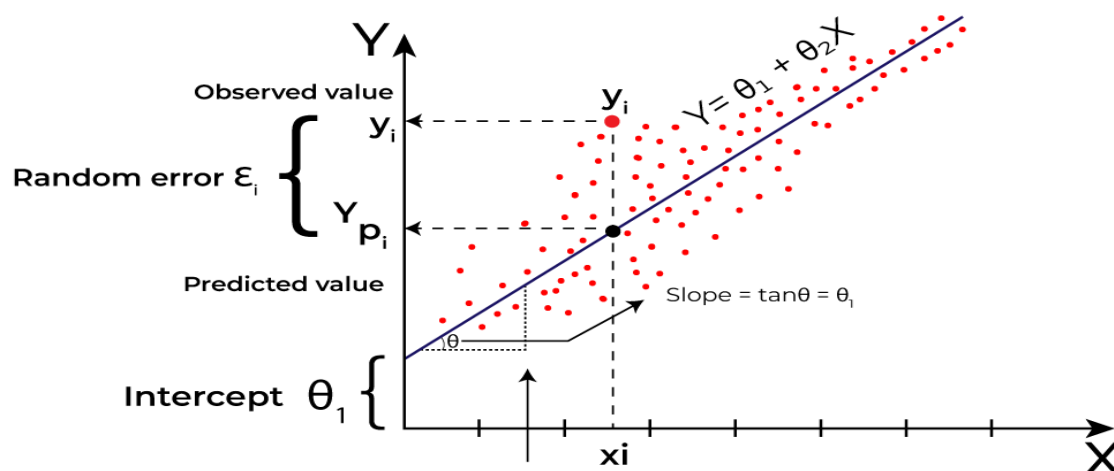
- The **mean**, **median**, and **mode** are all the same in a normal distribution.
- The **total area under the curve** equals 1, representing the total probability.
- It is widely used in statistics, finance, and natural sciences due to its occurrence in many natural and human-made phenomena.

Linear Regression

Linear regression is a statistical method in machine learning used to model the relationship between a dependent variable and one or more independent variables. It assumes a linear relationship between variables.

What is the Best Fit Line?

In **linear regression**, our main goal is to find the **best-fit line**. This line shows the relationship between the **independent variable(s)** (X) and the **dependent variable** (Y). The best-fit line is special because it has the **smallest error** between the predicted values and the actual values.



Equation of the Best-Fit Line

The equation of the line is simple and straight. It tells us how changes in the independent variable(s) affect the dependent variable. The **slope** of the line shows how much Y changes when X changes by one unit.

The equation of the **best-fit line** in linear regression is:

$$Y = mX + c$$

Where:

- **Y**: Dependent variable (output or target).
- **X**: Independent variable (input or predictor).
- **m**: Slope of the line (rate of change of Y with respect to X).
- **c**: Intercept (value of Y when X = 0).

For multiple features (multiple inputs), the equation becomes:

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$$

Where:

- **b₀**: Intercept.
- **b₁, b₂, ..., b_n**: Coefficients (slopes for each independent variable).
- **X₁, X₂, ..., X_n**: Independent variables.

This equation defines the **best-fit line** for predicting Y.

1. Simple Linear Regression

Simple Linear Regression is used to predict the relationship between **one independent variable (X)** and **one dependent variable (Y)**.

Equation:

$$Y = mX + c$$

Where:

- **Y**: Dependent variable (output).
- **X**: Independent variable (input).

- **m**: Slope of the line (how Y changes with X).
- **c**: Intercept (value of Y when X = 0).

Example:

Predicting a person's salary (**Y**) based on their years of experience (**X**).

2. Multiple Linear Regression

Multiple Linear Regression is used to predict the relationship between **one dependent variable (Y)** and **multiple independent variables (X₁, X₂, ..., X_n)**.

Equation:

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n$$

Where:

- **Y**: Dependent variable (output).
- **b₀**: Intercept (value of Y when all X variables are 0).
- **b₁, b₂, ..., b_n**: Coefficients (slopes of the independent variables).
- **X₁, X₂, ..., X_n**: Independent variables (inputs).

Example:

Predicting a house price (**Y**) based on features like square footage (**X₁**), number of bedrooms (**X₂**), and location score (**X₃**).

Mean Squared Error (MSE)

Mean Squared Error (MSE) is an evaluation metric that calculates the average of the squared differences between the actual and predicted values for all the data points. The difference is squared to ensure that negative and positive differences don't cancel each other out.

Formula:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where:

- n is the number of data points.
- y_i is the actual or observed value for the i^{th} data point.
- $\hat{y}_i$ is the predicted value for the i^{th} data point.

MSE is a way to quantify the accuracy of a model's predictions. MSE is sensitive to outliers as large errors contribute significantly to the overall score.

Mean Absolute Error (MAE) is an evaluation metric used to assess the accuracy of regression models. It calculates the average of the absolute differences between the predicted and actual values.

The formula for MAE is:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

Here,

- n is the number of observations
- Y_i represents the actual values.
- $\hat{y}_i$ represents the predicted values

Lower MAE value indicates better model performance. It is not sensitive to the outliers as we consider absolute differences.

